

Triphasic Interface Engineering with Metallic Sn/N, B Co-Doped Carbon Matrix for Boosting Reaction Kinetics and Cycling Stability in Lithium–Sulfur Batteries

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Lithium–sulfur batteries undergo solid–liquid–solid phase transitions based on a dissolution–deposition reaction mechanism. To effectively suppress the shuttling of soluble polysulfides, catalysts should be incorporated into the cathode to enhance both the adsorption and conversion processes. The formation of a triphasic interface among the catalyst, conductive material, and electrolyte plays a key role in facilitating these reactions. In this study, a composite catalyst (Sn@NBGNs-CNTs), consisting of metallic tin microparticles anchored on nitrogen and boron co-doped graphene nanosheets and partially exfoliated carbon nanotubes, is synthesized as a sulfur host via a simple method. This structure effectively anchors polysulfides at the interface and provided abundant active sites to accelerate redox reaction kinetics. As a result, it facilitated charge transfer and polysulfide transport at the interface, leading to an increase in the nucleation–growth rate constants of Li_2S as determined using the critical deposition voltage from the potentiostatic intermittent titration technique. Consequently, the electrode exhibits excellent cycling stability, retaining 93% of its initial capacity after 350 cycles at 1 C with an extremely low-capacity decay rate of 0.003% per cycle.

governed by the dissolution-deposition reaction mechanism of the active material.^[1–3] During the discharge process, solid-state S_8 is stepwise reduced to form long-chain lithium polysulfides (e.g., Li_2S_8 , Li_2S_6 , and Li_2S_4), which dissolve into the liquid electrolyte. As reduction continues, these polysulfides precipitate as solid-state $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, completing the discharge. Correspondingly, the reduced products are re-oxidized to lithium polysulfides and $\alpha/\beta\text{-S}_8$.^[4] Throughout this reaction pathway, the dissolved lithium polysulfides can shuttle to the lithium metal anode, leading to lithium passivation, increased electrolyte viscosity, and active material loss.^[5] Therefore, high cell performance requires effective suppression of the shuttle effect through strong adsorption and rapid conversion of polysulfides generated during the dissolution process. Furthermore, the theoretical capacity ratio between dissolution and

deposition processes is 1:3, highlighting the importance of effectively utilizing the deposition process.^[6] Therefore, a comprehensive electrochemical understanding of the interaction with polysulfides and the nucleation and growth processes is essential for the advancement of LSBs.

The introduction of a catalyst offers a potential strategy to address the aforementioned two challenges by forming a triphasic interface among the catalyst, electrolyte, and conductive material, which facilitates both the adsorption and conversion of polysulfides,^[7] as shown in **Figure 1a**. This interfacial environment can enhance the kinetic behavior of soluble polysulfides, thereby enabling appropriate nucleation and regulated growth of solid-phase Li_2S . Therefore, to introduce the aforementioned catalytic effects, a wide range of materials including oxides,^[8] nitrides,^[9] sulfides,^[10] and metals^[11] have been applied in LSBs. Among them, metal-based catalysts can serve as active centers due to their favorable electronic and chemical properties, thereby enhancing the redox kinetics of polysulfides.^[12] Tin (Sn), in particular, is a cost-effective material with excellent electrical conductivity and acid resistance.^[13,14] Its high valence p-band level endows it with oxophilic characteristics,^[14] which may facilitate charge transfer during the redox reactions of sulfur species that possess similar chemical behavior to oxygen. In the case of oxides

1. Introduction

Lithium-sulfur batteries (LSBs) undergo solid-liquid-solid phase transformations and polysulfide conversion-shuttling processes,

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DOI: 10.1002/sml.202503534

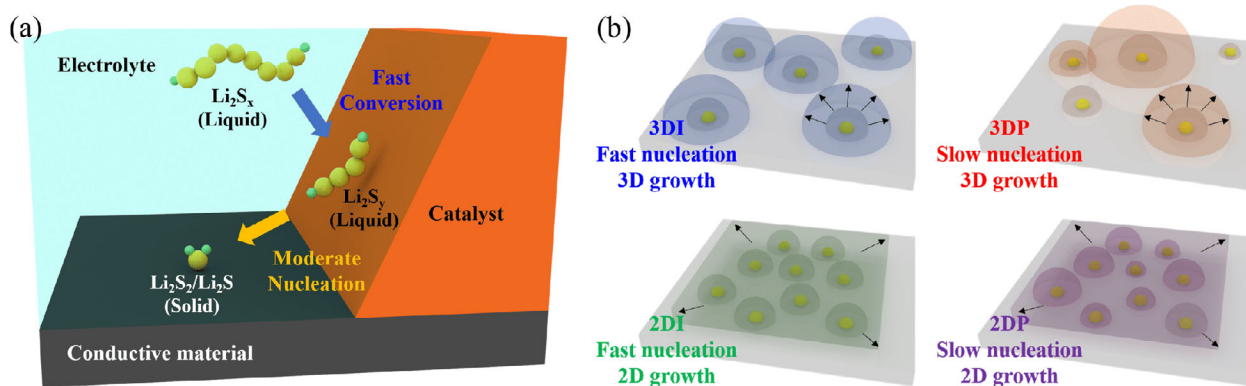


Figure 1. a) A schematic illustration of the role of catalysts in contributing to the redox reactions of polysulfides in LSBs, b) Schematic illustrations of electrochemical deposition models of nucleation and growth processes in LSBs.

such as SnO_2 , research has primarily focused on enhancing polysulfide adsorption through surface polarity, thereby suppressing the shuttle effect. In contrast, metallic catalysts such as Sn can improve sulfur utilization by accelerating the conversion of adsorbed polysulfides and promoting the reaction kinetics of the insulating Li_2S with their high electrical conductivity.^[15,16] When reviewing previous studies on metallic Sn catalysts, Marangon et al. achieved 800 mA h g^{-1} at 1 C by co-loading nanometallic Sn with sulfur, improving conductivity and reducing interfacial resistance.^[17] Jin et al. coated a separator with a microporous composite of N-doped carbon and Sn metal, enhancing polysulfide capture and improving the performance of LSBs.^[18] Wang et al. employed a metallic tin-modified nitrogen-doped carbon skeleton as an interlayer, achieving 98% coulombic efficiency after 400 cycles at 1 C .^[19] While these studies highlight the improved reaction kinetics enabled by Sn metal catalysts, their effect on sulfur deposition behavior in Li–S batteries remains insufficiently addressed.

This battery system occupies an intermediate domain between classical electrochemical deposition and conventional battery systems, enabling mutual influence between the two fields. This implies that theoretical frameworks and electrochemical analyses developed for classical electrodeposition can be applied to LSBs, particularly to describe the deposition of Li_2S during discharge process and the re-deposition of S_8 during charging process via polysulfides redox reactions. Therefore, a more comprehensive and integrated perspective, supported by rigorous electrochemical analysis, is essential for advancing fundamental insight into this complex electrochemical reaction. Recent studies have referenced classical electrochemical deposition theory, focusing on nucleation rate and growth dimensions, and have employed nucleation-growth models to interpret the deposition mechanism in LSBs.^[20,21] From the current-time response plot obtained through chronoamperometry, four classical electrochemical deposition models in Figure 1b can be applied in LSBs.^[20] Based on the nucleation rate, deposition can be categorized as either Instantaneous (I) or Progressive (P): i) In instantaneous nucleation, the nucleation rate is fast, leading to the rapid depletion of nucleation sites at the early stage. ii) In progressive nucleation, the nucleation rate is relatively slower.^[22] Based on the growth dimensions, deposition can be categorized into 2D planar growth and 3D hemispherical growth: i) Bewick, Fleis-

chman, and Thirsk (BFT) model describes the formation of a planar deposition through the incorporation of adjacent nuclei.^[23,24] ii) Scharifker-Hills (SH) model represents the growth process driven by 3D bulk diffusion.^[25,26] These electrochemical deposition models (2DI, 2DP, 3DI, and 3DP) enable the investigation of the nucleation-growth behavior of Li_2S using the chronoamperometry technique.^[11]

Using the previously described mathematical electrochemical deposition models, the nucleation and growth rate constants can be compared based on the time corresponding to the peak current in the potentiostatic curves.^[23,25] In general LSBs studies, chronoamperometry is applied at a specific identical apparent voltage, and the deposition capacity and the time corresponding to the peak current are compared and described.^[20,27,28] However, unlike the direct chronoamperometric analysis using a polysulfide solution, the potentiostatic intermittent titration technique (PITT) is used to determine the critical deposition potential for nucleation onset which can provide a more accurate interpretation of the nucleation-growth process in each system.^[29–31] Additionally, kinetics for concurrent nucleation and growth process can be obtained through the Johnson-Mehl-Avrami (JMA) equation.^[29,32] Based on this, the corresponding equation can be applied to calculate and compare the dimension of deposition process and the kinetic constants related to nucleation and growth rates. The dimensionality of deposition process can be used to determine the Avrami exponent.^[33]

In this study, metallic Sn microparticles anchored on nitrogen and boron co-doped graphene nanosheets and partially exfoliated carbon nanotube supports (Sn@NBGNs-CNTs) were synthesized and applied to lithium–sulfur batteries. This work aims to elucidate the enhancement of electrochemical reaction kinetics and the increased nucleation–growth rate constants during Li_2S deposition, attributed to the formation of a triphasic interface among the metal catalyst, carbon conductive materials, and electrolyte.

2. Results and Discussion

2.1. Materials Characterization

The synthesized Sn@NBGNs-CNTs composite structure, as observed in the scanning electron microscope (SEM) images

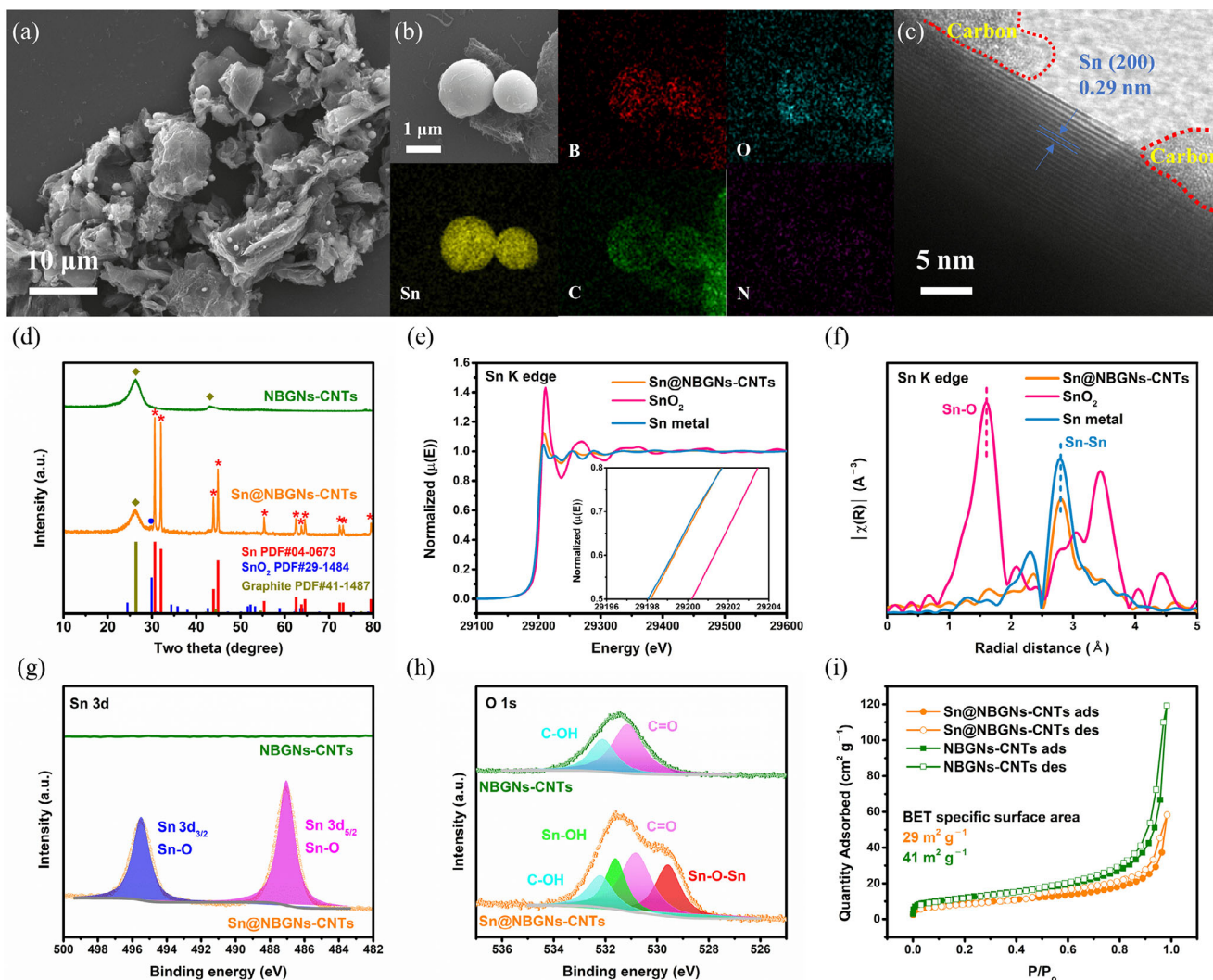


Figure 2. Materials characterization of Sn@NBGNs-CNTs and NBGNs-CNTs. a) SEM images of Sn@NBGNs-CNTs, b) high-resolution SEM images and EDS mapping of Sn@NBGNs-CNTs, c) high-resolution TEM images of Sn@NBGNs-CNTs, d) XRD patterns of two samples, e) Sn K edge XANES spectra of two samples, f) k^3 -weighted EXAFS spectra of two samples, g) XPS analysis of the Sn 3d spectra in two samples, h) XPS analysis of the O 1s spectra in two samples, i) N_2 sorption isotherms of two samples.

(Figure 2a,b), consists of multilayered wrinkled graphene nanosheets coexisting with carbon nanotubes and micron-sized metallic Sn particles formed by hydrogen reduction. The Sn mass fraction was confirmed to be 37.6 wt.% based on energy-dispersive X-ray spectroscopy (EDS) analysis (Table S1, Supporting Information). In contrast, the Sn-free sample (NBGNs-CNTs) consisted solely of carbon materials without any Sn microparticles (Figure S1, Supporting Information). As shown in Figure 2b,c, the spherical Sn particles are partially encapsulated by the N, B co-doped carbon structures. Additionally, Figure 2b and Figure S1 (Supporting Information) demonstrate uniform elemental distribution in both samples, as revealed by EDS mapping. From the transmission electron microscopy (TEM) image in Figure 2c, the observed interplanar spacing of 0.29 nm corresponds to the (200) crystallographic planes of metallic Sn. Moreover, electron energy loss spectroscopy (EELS) mapping revealed that the Sn particles predominantly consist of Sn with surface

oxidation, as indicated by the presence of oxygen (Figure S2, Supporting Information). TEM images and EELS mappings of the carbon matrix regions for both catalyst samples are provided in Figure S3 (Supporting Information). These results confirm the homogeneous distribution of nitrogen and boron, as well as the presence of partially exfoliated carbon nanotubes and widely spread graphene nanosheets.

The X-ray diffraction (XRD) pattern of Sn@NBGNs-CNTs corresponds to tetragonal metallic Sn phase (PDF#04-0673), orthorhombic SnO_2 phase (PDF#29-1484), and hexagonal graphite-2H phase (PDF#41-1487), as shown in Figure 2d, confirming the coexistence of catalytic and conductive carbon materials. The Sn phase shows high crystallinity, while SnO_2 exhibits only a faint (111) peak, suggesting a core-shell structure with a thin SnO_2 shell formed on the Sn surface, consistent with the TEM observations. This surface SnO_2 layer may serve as adsorption sites for lithium polysulfides via electrostatic interaction. Additionally,

a broad (002) peak at 26.2° is observed, attributed to the abrupt reduction of graphene oxide and partial exfoliation of CNT outer walls.^[34]

XAFS and XPS analyses further confirm the presence and oxidation state of Sn (Figure 2e–h). In Figure 2e, the XANES spectrum of Sn@NBGNs-CNTs closely matches that of metallic Sn, indicating a predominant Sn^0 oxidation state in the bulk. This indicates the oxidation state characteristic of Sn^0 , confirming that Sn in the sample predominantly possesses metallic properties in the bulk. Figure 2f shows that Sn–Sn bond distances (2.5–3.5 Å) are observed, while Sn–O distances (1.0–2.0 Å) appear only in the SnO_2 reference.^[35,36] The absence of Sn–O peaks in the Sn@NBGNs-CNTs sample suggests that Sn remains metallic in the bulk. However, the surface property exhibited the presence of SnO_2 due to surface oxidation, corresponding to the results shown in Figure S2 (Supporting Information). The Sn 3d spectrum of Sn@NBGNs-CNTs (Figure 2g) shows peaks at 495.5 and 487.1 eV, corresponding to $\text{Sn } 3d_{3/2}$ and $\text{Sn } 3d_{5/2}$, respectively, indicating a Sn^{4+} oxidation state and the formation of surface SnO_2 , without metallic Sn^0 .^[37] The O 1s spectrum of Sn@NBGNs-CNTs (Figure 2h) exhibits a distinct Sn–O–Sn (lattice O) peak at 529.6 eV, lower than the peaks for C=O/COO (530.9 eV), Sn–OH (531.7 eV), and C–OH/C–O–C (532.3 eV), due to its strong lattice binding energy.^[38,39] These results confirm the successful synthesis of a Sn-carbon composite with electrocatalytic functionality.

Nitrogen and boron in the carbon matrix is further confirmed by XPS (Figure S4, Supporting Information). The B 1s spectrum (Figure S4a, Supporting Information) is similarly deconvoluted into B–C, BC_2O , B–N, and BCO_2 peaks. The N 1s spectrum (Figure S4b, Supporting Information) is similarly deconvoluted into pyridinic-N (N-6), C–N–B, pyrrolic-N (N-5), and quaternary-N (N-Q), indicating that both B and N are incorporated as active sites.^[40] Raman shifts of the Sn@NBGNs-CNTs observed at 141 and 167 cm^{-1} correspond to substoichiometric SnO_x phase ($1 < x < 2$), 243 cm^{-1} to SnO phase (A_{1g}), and 633 cm^{-1} to SnO_2 (A_{1g}), confirming surface oxidation (Figure S5, Supporting Information).^[41–43] The I_D/I_G ratios of both samples are in the range of 1.1–1.2, indicating a high degree of disorder in the carbon structure due to N, B co-doping and defects.^[44] This structural disorder can provide effective adsorption sites for polysulfide conversion reactions. In addition, the minimal difference in I_D/I_G between the two samples suggests that Sn incorporation does not significantly influence the carbon disorder.

N_2 sorption isotherms (Figure 2i; Figure S6, Supporting Information) were analyzed using the Brunauer–Emmett–Teller (BET) method to determine the specific surface areas, and the Barrett–Joyner–Halenda (BJH) method for the pore distribution. Sn@NBGNs-CNTs exhibited a reduced specific surface area ($29\text{ m}^2\text{ g}^{-1}$) and pore volume ($0.089\text{ cm}^3\text{ g}^{-1}$), compared to NBGNs-CNTs ($41\text{ m}^2\text{ g}^{-1}$ and $0.171\text{ cm}^3\text{ g}^{-1}$), attributed to the presence of heavy Sn particles. The Sn content can be determined through thermogravimetric analysis (TGA), as shown Figure S7 (Supporting Information). For NBGNs-CNTs, 19.0 wt.% of BN residue remains under air conditions at 900°C in Figure S8a (Supporting Information).^[45] In Sn@NBGNs-CNTs, 53.7 wt.% of SnO_2 residue remains in Figure S8b (Supporting Information), corresponding to 32.1 wt.% Sn by stoichiometric back-calculation. This matches well with the EDS result (Table S1, Supporting Information), as stated in Note S1 (Supporting Information).

2.2. Comparison of Electrochemical Cell Performance Induced by the Triphasic Interface

To further evaluate the electrochemical performance of the as-assembled cells, galvanostatic charge–discharge (GCD) tests were conducted at a current density of 0.1 C. As shown in Figure 3a, two distinct voltage plateaus appear during both discharge and charge. The upper plateau corresponds to the reduction of elemental sulfur to long-chain polysulfides, while the lower plateau represents further reduction to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. Conversely, the charging process reflects oxidation from Li_2S to soluble polysulfides and subsequently to elemental sulfur. The Sn@NBGNs-CNTs electrode exhibited a higher discharge capacity of $1,061\text{ mA h g}^{-1}$ compared to 903 mA h g^{-1} for NBGNs-CNTs at 0.1 C, indicating improved sulfur utilization. Furthermore, Sn@NBGNs-CNTs exhibited a smaller voltage hysteresis of 130 mV compared to 135 mV for NBGNs-CNTs at 50% depth of discharge capacity, signifying enhanced reaction kinetics. Li_2S is intrinsically insulating, exhibiting low electrical conductivity and lithium diffusivity. Its strong binding energy and high standard formation enthalpy result in a high activation energy barrier.^[46] This leads to voltage dips observed during the onset of both deposition and activation, where Li_2S forms and reacts, respectively. As illustrated in Figure 3b, the polarization related to Li_2S deposition (blue region) and dissolution (pink region) was evaluated. The deposition voltage for Sn@NBGNs-CNTs was 2.092 V (vs 2.086 V for NBGNs-CNTs), and the dissolution voltage was 2.223 V (vs 2.235 V for NBGNs-CNTs), supporting the improved kinetics in both directions.

Rate capabilities were investigated at different current densities ranging from 0.1 C to 5 C rate under a sulfur loading density of 1.2 mg cm^{-2} and an electrolyte-to-sulfur (E/S) ratio of $25\text{ }\mu\text{L mg}^{-1}$ (Figure 3c). Sn@NBGNs-CNTs delivered specific capacities of 1061, 731, 647, 572, 505, 473, 451, and 397 mA h g^{-1} at C-rates of 0.1 C, 0.2 C, 0.5 C, 1 C, 2 C, 3 C, 4 C, and 5 C, respectively, significantly outperforming the capacities of 903, 619, 540, 490, 449, 408, 353, and 275 mA h g^{-1} obtained by NBGNs-CNTs. In the subsequent 100-cycles test at 0.5 C, the Sn@NBGNs-CNTs demonstrated a higher capacity retention of 88.5% than NBGNs-CNTs (83.4%). Evaluating rate performance is essential for demonstrating their capability to operate under high current conditions. The GCD profiles for various C-rates are presented in Figure S9 (Supporting Information). Moreover, the comparison of Li_2S deposition (yellow region) and dissolution (gray region) voltages across various rate conditions (Figure 3d) reveals that Sn@NBGNs-CNTs consistently exhibit lower overpotentials than NBGNs-CNTs, in agreement with in Figure 3b.

To estimate long-term cycling stability of the as-assembled cells, charging and discharging process was conducted for a continuous 350 and 500 cycles at a constant current of 1.0 C and 0.5 C, respectively. Figure 3e,f shows that Sn@NBGNs-CNTs achieved 93.0% retention over 350 cycles at 1.0 C and 77.1% retention over 500 cycles at 0.5 C, significantly outperforming NBGNs-CNTs (57.6% retention over 234 cycles at 1.0 C and 75.6% retention over 500 cycles at 0.5 C), experienced a short-circuit failure (Figure S10, Supporting Information). Sn@NBGNs-CNTs also demonstrated superior capacity stability, with decay rates of 0.003% per cycle (vs 0.374% for NBGNs-CNTs over 234 cycles) over 350 cycles at 1.0 C and

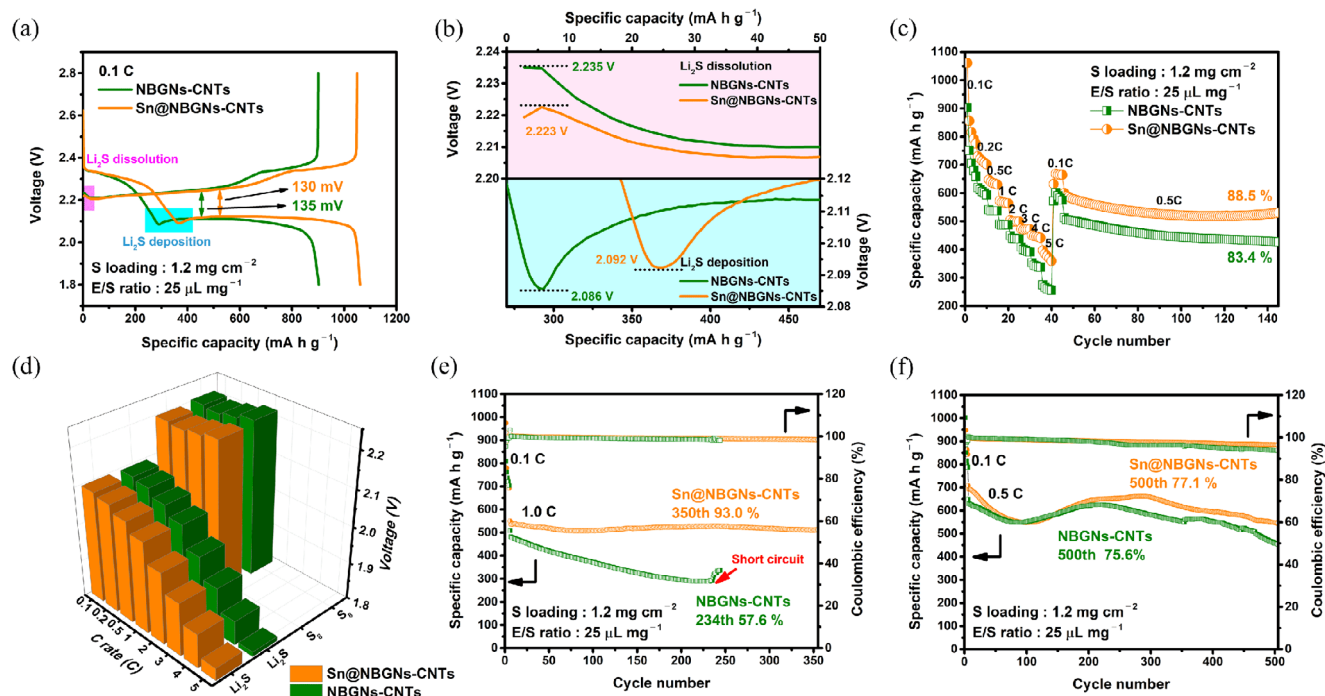


Figure 3. Electrochemical cell performance of Sn@NBGNs-CNTs and NBGNs-CNTs (Sulfur loading: 1.2 mg cm⁻², E/S ratio: 25 μ L mg⁻¹). a) Galvanostatic discharging/charging profiles at 0.1 C of both samples, b) Li₂S deposition and dissolution voltage in the highlighted regions of galvanostatic discharging/charging profiles, c) Rate capability tests of both samples, d) Li₂S deposition and dissolution polarization at different rates, e, f) Life cycling test at 1.0 C and 0.5 C of both samples.

0.09% per cycle (vs 0.234% for NBGNs-CNTs over 500 cycles) over 500 cycles at 0.5 C.

In Figure S11 (Supporting Information), in-situ XRD analysis was conducted to examine the differences in the reaction mechanisms and kinetic behaviors of LSBs^[47] arising from the triphasic interface. Both samples showed the α -S phase peak between 22° and 28° at the initial state. During discharge, sulfur dissolved and disappeared, and at a peak of 27° corresponding to the (111) plane of Li₂S emerged. Upon charging, this peak vanished, α -S reappeared at the final stage. Notably, the deposition of Li₂S occurred more rapidly in the presence of the triphasic interface, confirming the catalyst-induced enhancement in the nucleation-growth process. Therefore, it is necessary to compare how the kinetic behavior changes during the charge–discharge process by analyzing the deposition process.

2.3. Electrochemical Comparison of Diffusion-Controlled and Capacitive/Adsorptive Behaviors of Two Samples

The CV technique provides fundamental insights into the redox reactions in LSBs and enables differentiation between diffusion-controlled and capacitive/adsorption currents contributions, particularly within the Li₂S and S₈ deposition regions, as described in Note S2 (Supporting Information).^[48,49] As shown in Figure 4a, two pairs of redox peaks were observed during the discharge and charge, consistent with the GCD profiles in Figure 3a. The cathodic peaks correspond to the stepwise reduction of elemental sulfur to soluble polysulfides (C1), followed by further reduction

to solid Li₂S₂/Li₂S (C2). The anodic peaks indicate the oxidation of Li₂S to soluble polysulfides (A1), and subsequently to elemental sulfur (A2). Compared to NBGNs-CNTs, the Sn@NBGNs-CNTs exhibited lower polarization at the C2 and A2 peaks, which correspond to the Li₂S and S₈ deposition regions, as shown in Figure 4b. These results suggest that the composite interface between the metallic Sn and conductive carbon effectively lowers interfacial resistance during deposition reactions.

To distinguish between diffusion-controlled and capacitive contributions, CV measurements were conducted at scan rates ranging from 0.02 to 0.64 mV s⁻¹ (Figure 4c). According to the Randles–Sevcik equation, the peak current (i_p) is proportional to the square root of the scan rate.^[50] The peak current values increased linearly with the square root of the scan rate for all cathodic and anodic peaks, and the corresponding slopes were estimated in Figure 4d. In both regions, Sn@NBGNs-CNTs exhibited steeper slopes than NBGNs-CNTs, indicating enhanced mass transfer kinetics within the composite catalyst structure and conductive carbon. The overall current consists of two components: a diffusion-controlled current proportional to $v^{0.5}$ (Randles–Sevcik equation), and a capacitive–adsorptive current proportional to v .^[51] The slope for the A2 peak in Sn@NBGNs-CNTs was 0.83, higher than 0.78 for NBGNs-CNTs, suggesting a greater capacitive/adsorptive contribution for Sn@NBGNs-CNTs.

To evaluate the shuttle current arising from polysulfide migration between the cathode and anode in LSBs,^[52] the cells were held at 2.2 V and the resulting current was monitored over time, as presented in Figure 4f. Sn@NBGNs-CNTs exhibited a reduced

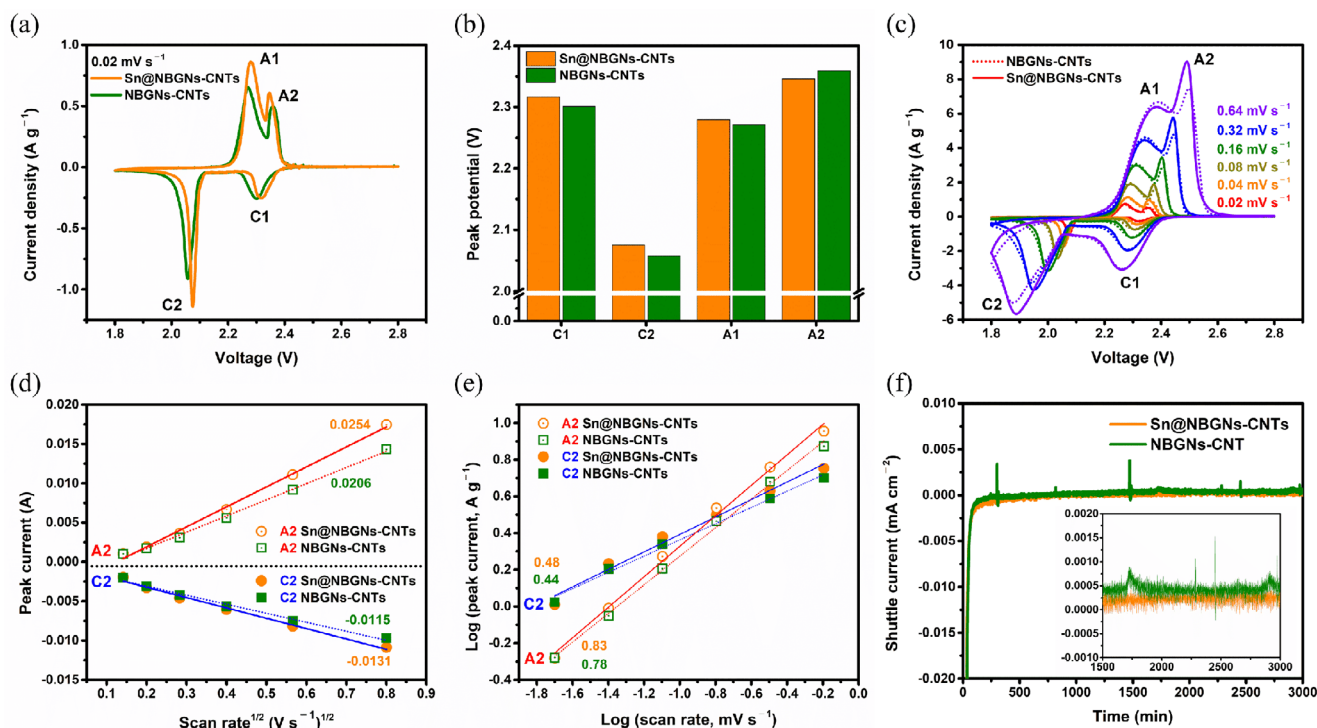


Figure 4. Electrochemical comparison of diffusion-controlled and capacitive/adsorptive behaviors of both samples. a) Cyclic voltammetry (CV) profiles of two samples at 0.02 mV s⁻¹, b) Corresponding peak potential of two samples in CV profiles, c) CV profiles of two samples at different scanning rates, d) The scanning rate-dependent peak currents plot at A2 and C2 peaks using Randles-Sevcik equation, e) The logarithmic slope value of the A2 and C2 peak calculated using the scan rate dependence equation, f) Shuttling current of two samples at 2.2 V.

shuttle current compared to NBGNs-CNTs, suggesting more effective suppression of the shuttle effect. This improvement is attributed to the synergistic adsorptive and catalytic properties of metallic Sn and the N, B co-doped carbon matrix. To further compare the polysulfides chemisorption, a 2 mM Li₂S₆ solution was prepared, and each catalyst was added in amounts normalized to a BET surface area of 5 m² (Figure 2i). After 2 h of immersion, UV-vis spectra were collected (Figure S12, Supporting Information). The Sn@NBGNs-CNTs sample showed a more pronounced decrease in absorbance in the 300–400 nm range, corresponding to polysulfide species. These findings are consistent with the reduced shuttle current shown in Figure 4f and support that the triphasic interface facilitates more effective polysulfide adsorption.

2.4. Electrochemical Comparison of Reaction Kinetics Affected by the Triphasic Interface

As shown in Figure 5a, GITT analysis was performed to evaluate the stepwise overpotentials and internal resistance within the cell. GITT enables the determination of overpotential at each step by comparing the quasi-equilibrium voltage (quasi-open circuit voltage, QOCV) with the closed-circuit voltage (CCV), which includes polarization.^[53] The overpotential is calculated as the difference between QOCV and CCV. For additional details, refer to Note S3 (Supporting Information). Based on the calculated internal resistance values presented in Figure 5b, the Sn@NBGNs-

CNTs exhibits lower resistance at the onset of both Li₂S deposition and dissolution. This result suggests that the triphasic interface facilitates both the formation and decomposition of electrically insulating Li₂S. To assess the evolution of voltage hysteresis over long-term cycling, GCD profiles at the 6th, 100th, and 200th cycles at 1C were analyzed, as shown in Figure 5c. At the 6th cycle, voltage hysteresis was nearly identical for both samples: 291 mV for Sn@NBGNs-CNTs and 293 mV for NBGNs-CNTs. However, after 100 cycles, the hysteresis values diverged to 238 and 266 mV, respectively, and further to 250 and 306 mV after 200 cycles, respectively. This increasing disparity indicates a difference in catalyst deactivation and degradation over long-term cycling, highlighting the superior cycling stability of Sn@NBGNs-CNTs.

To investigate the effect of the triphasic interface on interfacial charge-transfer reactions, electrochemical impedance spectroscopy (EIS) analysis was employed. EIS allows for the comparison of charge-transfer resistance (R_{ct}), which reflects the interfacial kinetics between the electrode materials and electrolyte.^[54] As shown in Figure 5d, the ex situ Nyquist plots obtained during the 1.0 C cycling test were fitted to an equivalent circuit model (Figure S13, Supporting Information) to extract R_{ct} values. After electrolyte infiltration and activation, Sn@NBGNs-CNTs exhibited a slightly higher resistance of 9.78 Ω compared to 9.37 Ω for NBGNs-CNTs. However, after 100 and 200 cycles, Sn@NBGNs-CNTs demonstrated significantly lower charge-transfer resistance values of 0.4 Ω (vs 1.29 Ω) and 1.25 Ω (vs 1.49 Ω), respectively (Figure 5e). This result is attributed to the facilitation of rapid interfacial charge transfer by the metallic Sn. Therefore, the

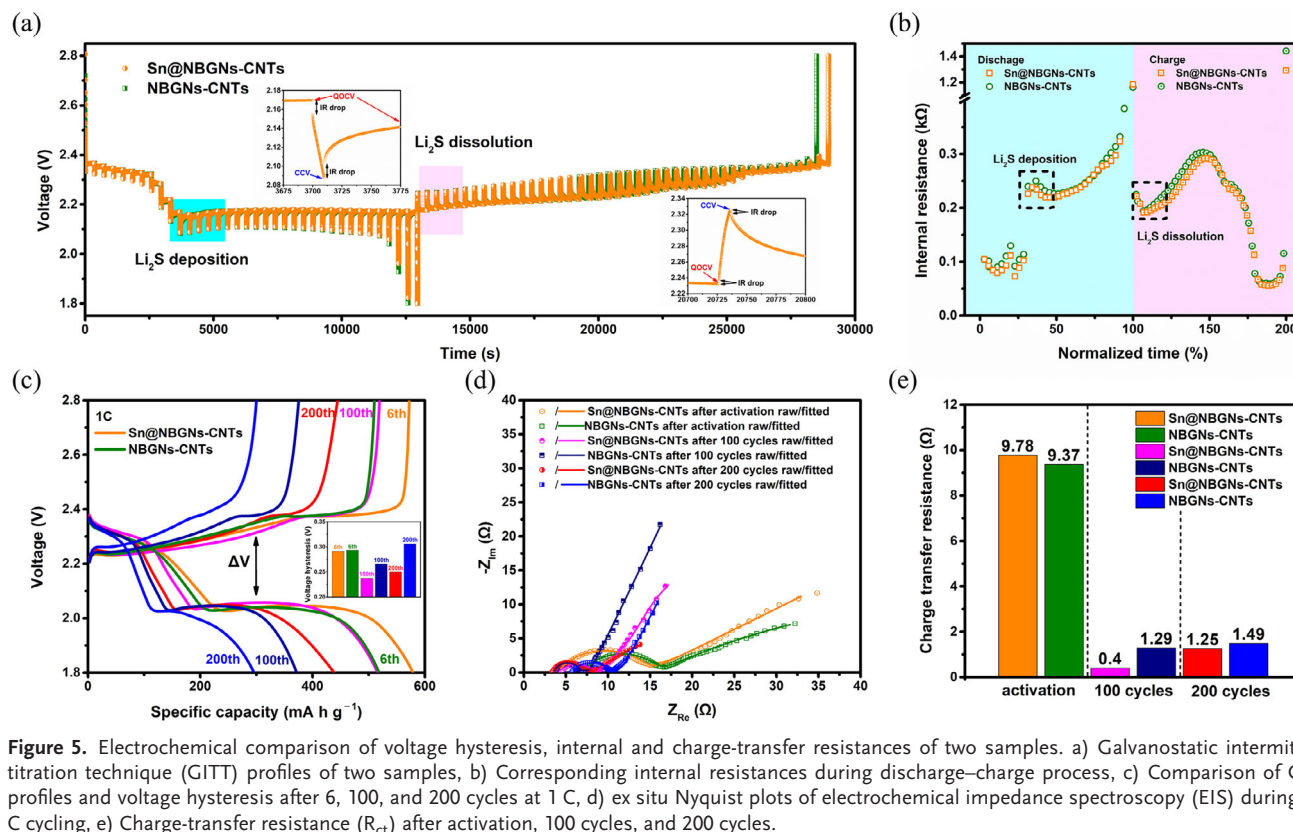


Figure 5. Electrochemical comparison of voltage hysteresis, internal and charge-transfer resistances of two samples. a) Galvanostatic intermittent titration technique (GITT) profiles of two samples, b) Corresponding internal resistances during discharge–charge process, c) Comparison of GCD profiles and voltage hysteresis after 6, 100, and 200 cycles at 1 C, d) ex situ Nyquist plots of electrochemical impedance spectroscopy (EIS) during 1.0 C cycling, e) Charge-transfer resistance (R_{ct}) after activation, 100 cycles and 200 cycles.

superior cycling stability of Sn@NBGNs-CNTs is attributed to its enhanced reaction kinetics and suppressed shuttle effect, resulting from strong polysulfide adsorption and rapid charge transfer at the triphasic interface.

2.5. Comparison of the Nucleation-Growth Morphology and Kinetics from Mathematical Modeling about Electrochemical Deposition Process

As previously mentioned, the deposition of Li_2S accounts for the $\approx 75\%$ of the total discharge capacity in LSBs, making it a critical factor in understanding the overall discharge behavior. Therefore, elucidating the deposition characteristics of insulating Li_2S is essential. PITT is a stepwise chronoamperometric technique that identifies the critical nucleation potential at which dissolved polysulfides overcome interfacial energy barriers to initiate nucleation.^[31] As shown in Figure S14 (Supporting Information), PITT was conducted using 10 mV increments across the discharge-charge process to investigate how the composite of metallic Sn and the carbon matrix affects the deposition process. The blue and pink regions correspond to the Li_2S deposition and dissolution reactions, respectively. In the magnified view of PITT in Figure 6a,b, the critical Li_2S deposition potential for Sn@NBGNs-CNTs is 2.114 V (vs 2.104 V for NBGNs-CNTs), indicating lower polarization of nucleation-growth process. Figure 6c shows the isolated chronoamperometric current response in the critical deposition region. The time to reach the maximum current (t_m) value is 7,642 s for Sn@NBGNs-CNTs, which is smaller

than 10,420 s for NBGNs-CNTs, indicating that Li_2S nucleation occurs more readily on the Sn@NBGNs-CNTs surface.^[48]

The nucleation and growth rate constants were applied using the JMA equation as described in Note S4 (Supporting Information).^[32] The current peak in the PITT-derived transients reflects the conversion reaction from polysulfides to solid Li_2S (Figure S15, Supporting Information). The initial monotonic current drop corresponds to the reduction of higher-order polysulfides (Li_2S_8 and Li_2S_6) to Li_2S_4 . This is followed by a current peak indicating the formation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. The charge under this peak represents the $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ conversion ratio, which is plotted over time and fitted using the JMA model (Figure 6d,e).^[55]

$$\xi = 1 - e^{-Bt^n} \quad (1)$$

where ξ is the Li_2S conversion fraction, n is the Avrami exponent (related to growth dimensionality), and B denotes the combined kinetic constant of nucleation and growth rates. The calculated n values are 3.3 for Sn@NBGNs-CNTs and 3.0 for NBGNs-CNTs, indicating predominantly 2D planar morphology. In addition, Sn@NBGNs-CNTs exhibited a higher B factor of 1.016×10^{-4} compared to 7.464×10^{-5} for NBGNs-CNTs, confirming that metallic Sn enhanced the deposition reaction kinetics through improved combined kinetic constants.

Figure 6f shows the Li_2S deposition behavior models using the BFT and the SH models as described in Note S5 (Supporting Information). In the early stages of t/t_m , both cases follow the

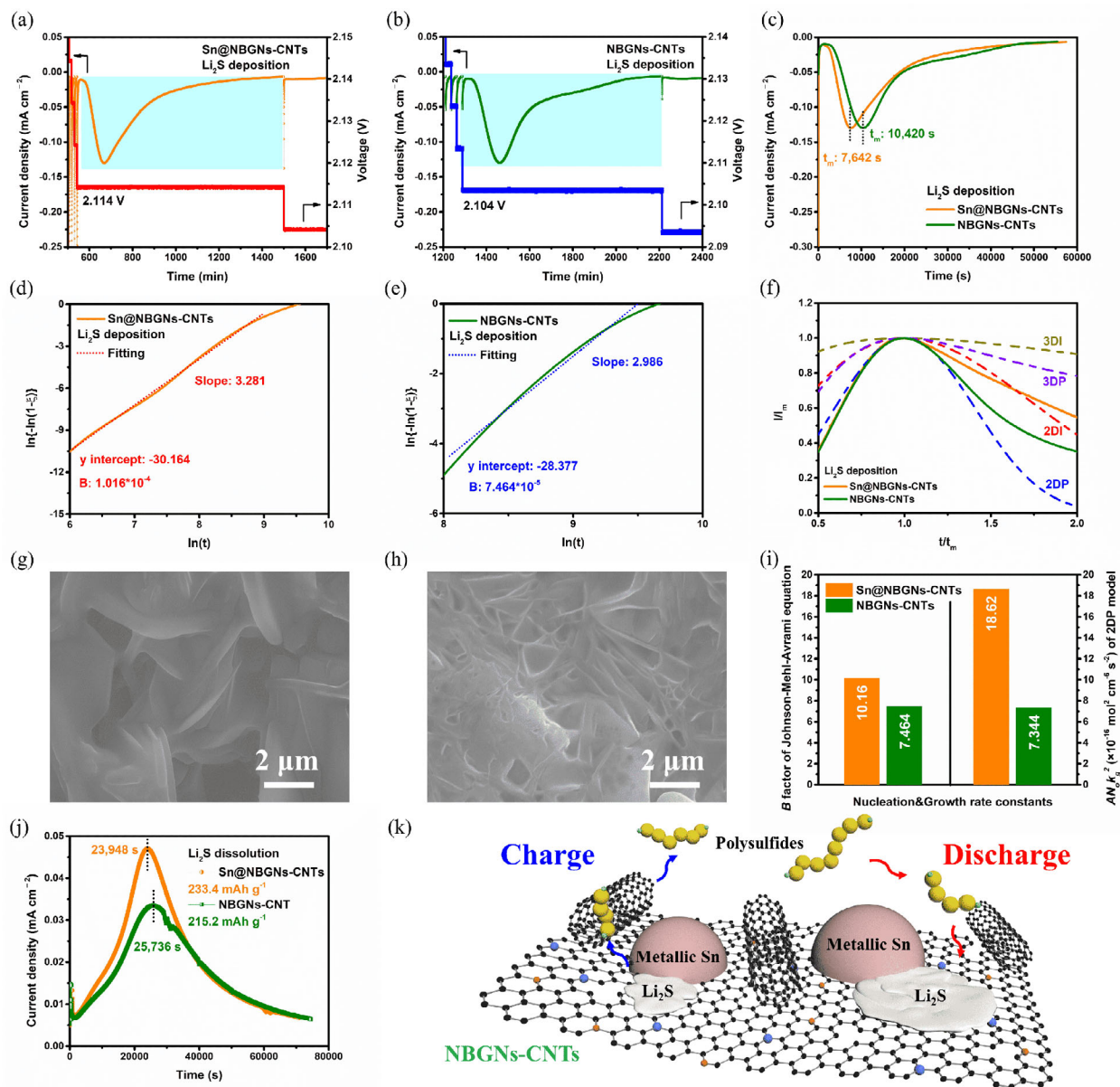


Figure 6. Interpretation of the Li₂S nucleation-growth process from PITT using the BFT and SH models, and the JMA equation. a, b) Magnified PITT plot of Sn@NBGNs-CNTs and NBGNs-CNTs, c) Current density–time transient profiles of independent Li₂S deposition regions derived from PITT, d, e) The fitting plot with kinetic factors from the JMA equation of Sn@NBGNs-CNTs and NBGNs-CNTs, f) The peak fitting with BFT and SH models from PITT, g, h) SEM images of the deposited Li₂S morphology of Sn@NBGNs-CNTs and NBGNs-CNTs, i) Nucleation and growth rate constants determined by two different methods, j) Current density–time transient profiles of independent Li₂S dissolution regions derived from PITT, k) Schematic illustration of the charge and discharge reaction processes in Sn@NBGNs-CNTs.

behavior of 2DP, allowing the comparison of ($AN_0k_g^2$ [mol² cm⁻⁶ s⁻³]) by applying $t_{m(2DP)}$:

$$t_{m(2DP)} = \left(\frac{2\rho^2}{\pi M^2 AN_0 k_g^2} \right)^{\frac{1}{3}} \quad (2)$$

Here, ρ and M are the density and molecular weight of deposited Li₂S, N_0 is the number density of active sites, and A and k_g are the nucleation and growth rate constants. Therefore,

the value of $AN_0k_g^2$ in the presence of the metallic Sn/carbon composite catalyst is 1.862×10^{-15} mol² cm⁻⁶ s⁻³, which is higher than 7.344×10^{-16} mol² cm⁻⁶ s⁻³ in its absence. This result is consistent with the B factor value and indicates that the triphasic interface contributes to the promotion of the nucleation and growth process. As shown in Figure 6g,h, SEM images illustrate the morphology of the deposited Li₂S on each catalyst surface. While both samples primarily exhibit a 2D planar deposition behavior, Sn@NBGNs-CNTs show a thicker layer, suggesting a tendency toward 3D growth, which is consistent

with the 3D fitting behavior observed in Figure 6d–f. From Figure 6i, the nucleation–growth kinetics were evaluated and compared using two approaches: 2D planar fitting based on the JMA equation and the BFT model. The results confirm that Sn@NBGNs-CNTs facilitated Li_2S deposition through the formation of a well-defined triphasic interface.

Figure 6j presents the dissolution behavior of Li_2S during the initial stage of the charging process, where activation and subsequent dissolution occur. The Sn@NBGNs-CNTs exhibited a shorter t_m of 23,948 s and a higher dissolution capacity of $233.4 \text{ mA h g}^{-1}$, compared to the t_m of 25,736 s and dissolution capacity of $215.2 \text{ mA h g}^{-1}$ observed for NBGNs-CNTs. These results demonstrate that the composite of metallic Sn and N, B co-doped carbon matrix facilitates 2D planar deposition of Li_2S and promotes the nucleation–growth process through the formation of a triphasic interface, thereby enhancing the deposition reaction kinetics and accelerating the dissolution of insulating Li_2S .^[56] Figure 6k provides a simplified illustration of the reaction mechanism in which, during the discharge process, soluble polysulfides undergo 2D planar Li_2S deposition at the interface between metallic Sn and NBGNs-CNTs, while in the subsequent charge process, Li_2S dissolves and converts back into polysulfides. Therefore, the introduction of Sn@NBGNs-CNTs forms a triphasic interface that enhances the kinetics of Li_2S deposition and dissolution, promotes the conversion reactions, and ultimately leads to improved electrochemical performance.

3. Conclusion

In this study, metallic Sn microparticles anchored on nitrogen and boron co-doped graphene nanosheets and partially exfoliated carbon nanotube supports were synthesized to construct a triphasic interface and introduced into LSBs with the aim of enhancing reaction kinetics and promoting the nucleation–growth process. The Sn@NBGNs-CNTs delivered an initial discharge capacity of $1,061 \text{ mA h g}^{-1}$ at 0.1 C and maintained a high-rate performance of 397 mA h g^{-1} at 5 C. Furthermore, after 350 cycles at 1 C, the electrode exhibited excellent cycling stability with 93% capacity retention and a capacity decay rate of only 0.003% per cycle, outperforming the counterpart without metallic Sn. The reduced voltage hysteresis and lower peak potentials observed in the deposition region indicate decreased overpotential, which is consistent with the reduced charge-transfer resistance and internal cell resistance. Additionally, by applying the JMA equation and BFT/SH models to the critical nucleation potential derived from PITT analysis, a 2D deposition morphology and relatively high nucleation–growth rate constants were identified. These findings confirm that the engineered interface between the metal catalyst and conductive matrix promotes the adsorption and conversion of polysulfides, as evidenced by reductions in overpotential, resistance, and improvements in nucleation–growth kinetic parameters. Therefore, the introduction of triphasic interface based on metal catalysts presents a promising design strategy for high-performance cathodes in LSBs.

4. Experimental Section

Detailed experimental procedures can be found in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

G.H.P. and S.R.M. contributed equally to this work. This research was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (RS-2024-00355916), by NRF grant funded by the Korea government(MSIT) (No. RS-2021-NR060090), and by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (RS-2024-00419413, HRD Program for Industrial Innovation).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

deposition, lithium-sulfur batteries, metallic Sn, nucleation-growth, triphasic interface

Received: May 16, 2025

Revised: June 17, 2025

Published online:

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